



Advanced Mathematical Economics **First Order Difference** **Equations (Numerical Problem)**

Dr. Hafiz Muhammad Qasim

Assistant Prof. Department of Economics

Akhuwat University

Hmq.economist@gmail.com

Outlines

- Summery of Previous Lecture
- Numerical Example



Summary of Previous Lecture

- In previous lecture we have derive the following general and definite solutions of first order difference equations:
- Standard form of first order difference equation
- $Y_{t+1} + aY_t = C$
- Non-Homogeneous Case if $C \neq 0$
- $Y_{t+1} + aY_t = C$
- Homogeneous Case if $C=0$
- $Y_{t+1} + aY_t = 0$

Summary of Previous Lecture

- The objective of difference equations is to find the time path:
- $Y_t = Y_c + Y_p \dots \dots \dots (4)$
- Particular Solution or equilibrium solution (Y_p)
- $Y_p = \frac{c}{(1+a)} \dots \dots \dots (8)$
- Complementary faction (Y_c)
- $Y_c = A(-a)^t \dots \dots \dots (10)$

Summary of Previous Lecture

- Obtaining General Solution; if $a \neq 0$

- $Y_t = A(-a)^t + \frac{C}{(1+a)} \dots \dots (11)$

- Definite Solution; $a \neq -1$

- $A = Y_0 - \frac{C}{(1+a)} \dots \dots (12)$ if $t=0$

- $Y_t = \{Y_0 - \frac{C}{(1+a)}\}(-a)^t + \frac{C}{(1+a)} \dots \dots (13)$

Summary of Previous Lecture

- General Solution ; $a = -1$
- $Y_t = A + Ct \dots\dots (14)$
- Definite Solution; if $a = -1$
- $A = Y_0 \dots\dots (15)$ assuming $t = 0$
- $Y_t = Y_0 + Ct \dots\dots (16)$

Example 04: Page 550

- Q: Solve the following first-order difference equation
- $Y_{t+1} - 5Y_t = 1; (Y_0 = \frac{7}{4})$
- $Y_{t+1} + aY_t = C$ (By comparing with standard form)
- $a=-5$ & $c=1$
- $Y_c = A(-a)^t \dots \dots \dots (10)$
- $Y_c = A(-(-5))^t$ by substituting the value of $(-a)$
- $Y_c = A(5)^t$

For Particular Solution

- $Y_p = \frac{c}{(1+a)} \dots\dots\dots(8)$
- $Y_p = \frac{1}{(1+-5)} \text{ (by substituting } c=1, \text{ and } a=-5)$
- $Y_p = -\frac{1}{4}$

For General Solution

- $Y_t = Y_c + Y_p \dots \dots (4)$

- $Y_t = A(-a)^t + \frac{C}{(1+a)} \dots \dots (11)$

- $Y_t = A(5)^t - \frac{1}{4}$

For Definite Solution

- Assume $t=0$
- $Y_t = A(5)^t - \frac{1}{4}$
- $Y_0 = A(5)^0 - \frac{1}{4}$ (by putting $t=0$)
- ($Y_0 = \frac{7}{4}$); (as we know, Y_0 given)
- $\frac{7}{4} = A(5)^0 - \frac{1}{4}$; (solving for A)
- $\frac{7}{4} = A(1) - \frac{1}{4}$; because $(5)^0=1$
- $A = \frac{7}{4} + \frac{1}{4} = \left(\frac{7+1}{4}\right) = \frac{8}{4}=2$
- $A=2$

Definite Solution

- $Y_t = \{Y_0 - \frac{c}{(1+a)}\}(-a)^t + \frac{c}{(1+a)} \dots\dots (13)$
- $Y_t = 2(5)^t + \frac{1}{4} ; (\textit{Definite Solution})$

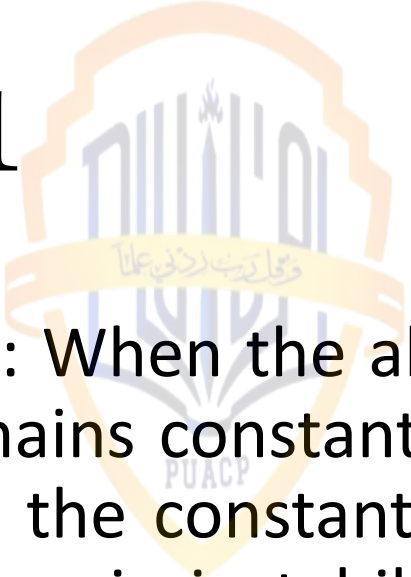
Dynamic Stability



- The dynamic stability of difference equations depends on the absolute value of exponential growth component; $b^t = (-a)^t$
- If $|b| > 1$ In this case, the absolute value of b is greater than 1, which means that the exponential term b^t grows rapidly as t increases. This leads to unbounded growth of Y_t over time, indicating dynamic instability.

$$|b| < 1$$

- If $|b| < 1$: Conversely, if the absolute value of b is less than 1, the exponential term b^t decreases rapidly as t increases. In this scenario, $Y_t \rightarrow 0$ approaches zero over time, indicating dynamic stability.


$$|b| = 1$$

- If $|b| = 1$: When the absolute value of b equals 1, the exponential term b^t remains constant over time. In this case, the behavior of Y_t depends on the constant term C . If $C \neq 0$, Y_t does not converge and exhibits dynamic instability. If $C=0$, the sequence remains constant over time, indicating dynamic stability.



•**Thank You**